

MATSUSHIMA AB, Japan

WMO: 475910

Lat: 38.403N

Lon: 141.212E

Elev: 5

StdP: 101.26

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.2	-4.1	-10.2	1.6	-1.7	-9.2	1.7	-1.0	13.7	2.4	12.6	2.5	2.7	300	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.6	29.9	24.5	28.1	24.2	27.1	23.8	25.9	28.1	25.2	27.4	24.6	26.5	4.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.2	27.2	24.2	19.2	26.4	23.9	18.8	26.1	79.9	28.5	76.5	27.2	73.8	26.6	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.3	9.0	DB	-8.9	33.1	1.8	1.7	-10.2	34.3	-11.3	35.3	-12.4	36.2	-13.7	37.4	(4)
(5)			WB	-9.6	26.6	1.7	0.9	-10.8	27.3	-11.8	27.8	-12.7	28.3	-13.9	29.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	0.8	1.4	4.4	9.5	14.8	18.5	22.5	24.0	20.7	14.9	8.6	3.3	(6)
(7)		DBStd	8.52	2.43	2.54	2.79	3.14	2.77	2.39	2.72	2.42	2.95	3.01	3.20	2.74	(7)
(8)		HDD10.0	1021	284	240	176	46	1	0	0	0	0	2	64	208	(8)
(9)		HDD18.3	2737	542	473	433	264	113	26	2	0	11	113	293	466	(9)
(10)		CDD10.0	1759	0	0	1	32	150	256	388	435	322	153	21	1	(10)
(11)		CDD18.3	433	0	0	0	0	4	32	132	177	82	6	0	0	(11)
(12)		CDH23.3	2071	0	0	0	1	16	60	598	1051	336	10	0	0	(12)
(13)		CDH26.7	366	0	0	0	0	2	6	116	204	38	1	0	0	(13)
(14)	Wind	WSAvg	3.8	4.4	4.5	4.6	4.3	3.8	3.3	3.0	3.1	3.2	3.4	3.6	4.1	(14)
(15)	Precipitation	PrecAvg	1153	41	37	69	90	101	126	156	125	161	128	73	47	(15)
(16)		PrecMax	1528	136	88	163	166	204	266	327	457	474	375	155	194	(16)
(17)		PrecMin	885	7	6	6	8	35	63	53	21	13	23	13	15	(17)
(18)		PrecStd	169	32	21	38	46	42	45	77	102	98	86	43	36	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	11.9	15.9	21.2	25.5	27.2	32.0	32.8	29.2	24.8	18.9	14.0	(19)
(20)			MCWB	6.6	7.1	8.9	12.5	17.3	20.2	24.7	24.7	23.4	18.7	13.7	10.9	(20)
(21)		2%	DB	8.0	9.2	13.0	18.2	22.2	25.0	29.2	30.1	27.8	22.8	17.0	11.8	(21)
(22)			MCWB	4.7	5.6	8.2	12.0	15.5	20.2	24.3	24.9	23.7	18.6	13.7	8.5	(22)
(23)		5%	DB	6.8	7.9	11.1	16.2	20.8	23.2	27.9	28.9	26.2	21.1	15.9	9.9	(23)
(24)			MCWB	3.6	4.6	7.2	11.3	15.2	19.3	23.9	24.7	23.1	17.6	12.7	6.8	(24)
(25)		10%	DB	5.2	6.2	9.9	14.9	19.0	22.1	26.2	27.8	25.1	20.0	14.8	8.2	(25)
(26)			MCWB	2.2	3.0	6.5	10.7	15.1	19.0	23.4	24.6	22.5	16.6	11.6	5.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	8.5	11.0	15.4	18.7	22.6	26.3	26.7	25.9	21.5	15.7	11.7	(27)
(28)			MCDWB	8.6	10.4	14.1	18.0	22.1	25.4	29.3	29.2	27.8	22.9	17.2	13.4	(28)
(29)		2%	WB	5.0	6.3	9.4	13.7	17.6	21.4	25.3	26.1	25.0	19.6	14.4	8.9	(29)
(30)			MCDWB	7.2	8.7	11.5	16.5	20.3	23.6	28.1	28.4	26.5	21.4	16.3	10.9	(30)
(31)		5%	WB	3.7	4.7	8.1	12.6	16.7	20.5	24.6	25.6	24.2	18.4	13.1	7.0	(31)
(32)			MCDWB	6.0	7.2	10.4	15.2	19.1	22.4	27.0	27.7	25.8	20.7	15.3	9.3	(32)
(33)		10%	WB	2.6	3.3	6.8	11.5	15.7	19.6	23.8	25.0	23.2	17.2	11.7	5.4	(33)
(34)			MCDWB	5.0	5.9	9.2	14.0	18.0	21.5	25.7	27.0	24.9	19.6	14.1	7.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.6	8.2	9.0	9.1	7.4	5.8	5.3	5.6	6.6	8.4	8.9	7.7	(35)
(36)			MCDBR	9.3	10.2	11.0	11.3	10.1	7.8	7.8	7.5	7.2	8.8	10.4	9.7	(36)
(37)			MCWBR	6.9	7.4	7.7	6.8	5.4	4.4	4.0	4.0	4.3	5.8	7.7	7.2	(37)
(38)		5% WB	MCDWR	9.0	9.8	10.1	9.4	7.4	6.3	6.4	6.0	5.6	7.2	9.6	9.7	(38)
(39)			MCWBR	7.1	7.8	8.1	6.4	4.8	4.1	3.7	3.6	3.9	5.7	7.9	7.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.369	0.414	0.488	0.537	0.558	0.552	0.561	0.547	0.480	0.440	0.401	0.377	(40)	
(41)		taud	2.171	2.045	1.896	1.839	1.851	1.944	1.983	2.011	2.163	2.175	2.188	2.183	(41)	
(42)		Ebn at Noon	777	787	763	754	747	751	740	739	767	755	735	735	(42)	
(43)		Edh at Noon	109	141	180	203	204	186	178	169	137	123	107	101	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.07	2.83	3.75	4.67	5.04	4.87	4.43	4.39	3.64	2.97	2.20	1.77	(44)	
(45)		RadStd	0.12	0.19	0.23	0.45	0.49	0.30	0.67	0.56	0.33	0.30	0.14	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.37	N/A	N/A	N/A	N/A

Nomenclature: See separate page